

DAY 04 – Excel & Business Analyst Notes

Focus: Lookup functions, INDEX/MATCH, data quality, validation, sorting/filtering, text cleaning, dates, and CRM analysis.

1. LOOKUP CONCEPT

- A lookup searches for a key/value and returns the corresponding value from another column or array.
- LOOKUP is an older function and generally uses approximate/last-applicable matching behavior.
- XLOOKUP is the modern lookup function and defaults to exact matching.
- INDEX + MATCH is a flexible combination for retrieving a value based on position.

2. XLOOKUP

- Syntax: =XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode])
- lookup_value = value to find.
- lookup_array = column/range where you search.
- return_array = column/range from which you return the result.
- if_not_found = optional message/value when no match exists.
- match_mode = optional; 0 means exact match. XLOOKUP already uses exact matching by default.
- Example: =XLOOKUP(A2, Customer_Master[Customer ID], Customer_Master[Country], "Not Found")
- If there is no match, "Not Found" is returned instead of #N/A.

3. IMPORTANT XLOOKUP ERROR TO AVOID

- Correct argument order: return_array → if_not_found → match_mode.
- Correct: =XLOOKUP(A2, Customer_Master[Customer ID], Customer_Master[Country], "Not Found", 0)

4. INDEX + MATCH

- MATCH = WHERE is it? INDEX = WHAT value is there?
- MATCH returns the position within the lookup array, not necessarily the worksheet row number.
- Example: =MATCH("C003", Customer_Master[Customer ID], 0)
- Example: =INDEX(Customer_Master[Country], 3)
- Combined: =INDEX(Customer_Master[Country], MATCH(A2, Customer_Master[Customer ID], 0))

- INDEX return array and MATCH lookup array should correspond to the same source table and ordering.

5. DATA QUALITY – DUPLICATES

- Never delete duplicate-looking records immediately.
- First identify the data grain: customer-level, transaction-level, etc.
- Understand the business key/primary identifier.
- Repeated Customer IDs can be valid when the grain is transaction-level.
- Take a backup before destructive cleaning.
- Check Transaction ID, Customer ID, Date, Product/Service and Amount.
- Use Conditional Formatting to identify possible duplicates.
- Use Remove Duplicates only after confirming the business rule.

6. DATA VALIDATION

- Data Validation controls what users can enter.
- CRM Status example: New | In Progress | Won | Lost.
- Excel path: Data → Data Validation → Allow: List.

7. SORTING VS FILTERING

- Filter temporarily displays only records matching a condition, such as Country = Kenya. Other records are hidden, not deleted.
- Sorting changes the order of records.
- Never independently sort one column in a multi-column raw dataset; it can break the relationship between fields. Sort the complete table/record set together.

8. TEXT CLEANING

- TRIM() removes unnecessary spaces.
- UPPER() converts text to uppercase.
- LOWER() converts text to lowercase.
- PROPER() converts text to proper case.
- Common formula: =TRIM(PROPER(A2))
- Example: " rahul sharma " → "Rahul Sharma"
- PROPER() can alter legitimate capitalization, so use it thoughtfully.

9. DATE HANDLING

- Valid Excel dates are generally stored internally as serial numbers and displayed using date formatting.
- =ISNUMBER(A2) helps check whether a value is numeric.
- =DAY(A2) returns the day number.

- =MONTH(A2) returns the month number.
- =YEAR(A2) returns the year.
- =TEXT(A2,"mmmm") returns the full month name.
- =TEXT(A2,"mmmm yyyy") returns month + year.
- Example: 25-Dec-2026 → DAY 25, MONTH 12, YEAR 2026.
- For December 2026: =TEXT(A2,"mmmm yyyy")
- Do not use TEXT(MONTH(A2),"mmmm") to get the month name; MONTH(A2) returns a number.

10. CRM LOOKUP EXERCISE

- XLOOKUP: =XLOOKUP([@[Customer ID]],Customer_Master[Customer ID],Customer_Master[Customer Name],"Customer Not Found")
- INDEX/MATCH: =IFERROR(INDEX(Customer_Master[Customer Name],MATCH([@[Customer ID]],Customer_Master[Customer ID],0)),"Not Available")

11. CRM SALES ANALYSIS

- For total Won Sales in Kenya:
=SUMIFS(Table1[Sales],Table1[Country],"Kenya",Table1[Status],"Won")
- Day 04 assessment dataset: Kenya Won Sales = 50,000 + 40,000 + 25,000 = 115,000.

12. BUSINESS ANALYST THINKING

- A single sales number is not enough to conclude that a country is performing well.
- Compare Actual Sales against Target.
- Calculate Achievement % and Variance.
- Understand the time period.
- Break performance down by Branch, Product/Service, Country and Salesperson.
- Verify measures and data quality.
- Investigate the ground-level cause before making a management conclusion.

13. KEY TAKEAWAYS

- XLOOKUP is generally preferred for modern Excel lookups and defaults to exact matching.
- MATCH tells you the position; INDEX retrieves the value.
- Customer ID repeated in transaction-level data is not automatically a duplicate.
- Transaction ID is important when investigating duplicate transactions.
- Use Data Validation for controlled input lists.
- Use Filter to temporarily hide nonmatching records.
- Sort the complete table, not an isolated column.
- Use TRIM/PROPER carefully for text standardization.
- Use DAY, MONTH, YEAR and TEXT for date analysis.
- Use SUMIFS for multi-condition business totals.

- Combine numbers with targets, time period and dimensions before drawing business conclusions.

14. REVISION POINTS

- XLOOKUP argument order: lookup_value, lookup_array, return_array, if_not_found, match_mode.
- INDEX/MATCH must use matching source arrays.
- MATCH returns position within the lookup array.
- Reference the actual date cell in TEXT(), e.g. =TEXT(A2,"mmm yy").